

ChiralFold: systematic detection of D-amino acid stereochemistry errors in the Protein Data Bank

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Abstract

Motivation: D-amino acids are essential in therapeutics and natural products, yet standard validators do not audit D-residue stereochemistry in deposited structures, and diffusion predictors violate D-chirality at high rates.

Results: We audited 12,573 D-residues in 4,616 PDB files and found 29 D-label/L-coordinate mismatches (0.23%) in 16 structures not covered by MolProbity's L-residue C α check. ChiralFold, calibrated on 362 structures (Ramachandran Spearman $\rho = 0.52$ vs wwPDB), detects errors via signed C α volume; Lean 4 proofs show chirality sign cannot factor through pairwise distances alone.

Availability: <https://github.com/Tommaso-R-Marena/ChiralFold>; `pip install chiralfold`; Python 3.9+; Linux/macOS/Windows (RDKit required).

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Supplementary information: Supplementary data and extended analyses are available at Bioinformatics online and at <https://github.com/Tommaso-R-Marena/ChiralFold/tree/master/paper/submission/bioinformatics>.

Keywords D-amino acid, chirality, Protein Data Bank, structure validation, AlphaFold 3, stereochemistry

1. Introduction

D-amino acids resist proteolysis and underpin mirror-image therapeutics, NRPS natural products, and bacterial cell-wall chemistry. AlphaFold 3 (AF3) produces a 51% per-residue chirality violation rate on D-peptides (Childs et al., 2025), and MolProbity—the standard PDB validator (Chen et al., 2010)—does not extend C α chirality checks to D-labeled residues. Deposited D-amino acid coordinates may therefore carry annotation errors that persist undetected.

We present ChiralFold, a pip-installable stereochemistry toolkit, and the first systematic PDB-wide audit of D-residue chirality. We classify error modes, benchmark the auditor against wwPDB/MolProbity, formalize a no-go theorem for distance-based chirality in Lean 4, and release reproducible code (v3.4.0).

2. Materials and methods

2.1. Signed tetrahedron volume

Chirality at C α uses the signed volume of the tetrahedron formed by backbone atoms N, C α , C, and C β :

$$V = (\mathbf{N} - \mathbf{C}_\alpha) \cdot [(\mathbf{C} - \mathbf{C}_\alpha) \times (\mathbf{C}_\beta - \mathbf{C}_\alpha)]. \quad (1)$$

For correctly deposited D-residues, $V < 0$ by CIP convention; $V > 0$ at a D-label indicates L-coordinates. The PDB survey used an independent numpy-only script (no ChiralFold imports). mmCIF-only entries ($n = 245$) were excluded from legacy-PDB coverage statistics.

2.2. ChiralFold auditor

The auditor scores C α chirality, bond geometry, hybrid Ramachandran grids, peptide planarity, steric clashes, and a composite 0–100 score. Ramachandran calibration used an era-representative stratified sample of 362 standard-protein structures (seed 20260612), comparing outlier percentages to wwPDB validation XML; one non-protein entry (5M2K) was excluded under a pre-specified criterion (`wwpdb_rama_outlier_pct > 50%`). See Supplementary Methods for bin definitions and chain-ID remapping.

2.3. Formal verification

Lean 4 proofs (`formal/chirality_nogo/`) show that reflection preserves pairwise distances but negates oriented volume, and that any model factoring through the distance matrix cannot represent signed chirality. AF3 correction reflects violating C α atoms across the N–C–C β plane; mirror-image and diastereomer modules are described in Supplementary Methods.

3. Results

3.1. PDB D-residue survey

Of 12,573 checkable D-residues, 29 (0.23%) in 16 structures showed positive signed volume despite D-labels (Fig. 1). Cross-referencing deposition metadata and literature yields six genuine stereochemistry errors (biology requires D, coordinates show

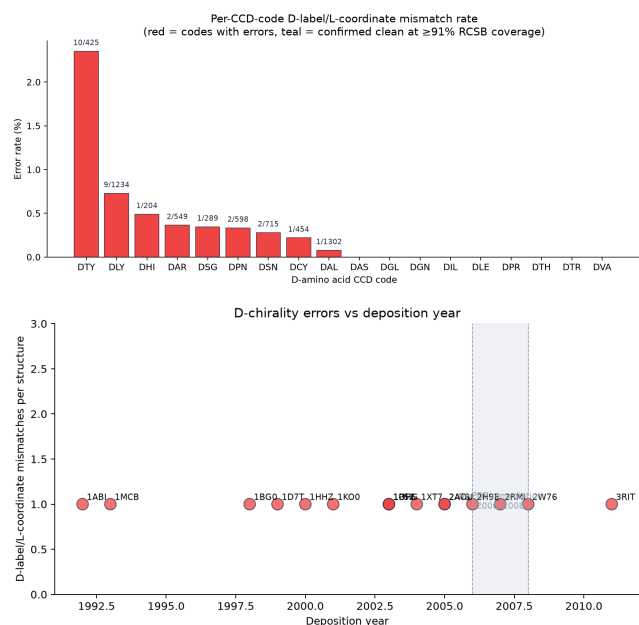


Figure 1 (a) Error rate per CCD code with Wilson 95% CIs; red: error-prone codes. (b) Deposition year vs error indicator; shaded band: 2006–2008 remediation.

L), 18 CCD-code misassignments across five structures (L-ligand labeled with D-code), three polymer mislabels, and two borderline cases (Supplementary Table S1). All 29 mismatches map to nine CCD codes; nine codes are confirmed clean at zero errors (Fisher's exact $p = 1.96 \times 10^{-10}$); DTY and DLY alone account for 19 errors. MolProbity's standard $C\alpha$ chirality check applies to L-residues only and does not flag these D-label mismatches. Thirteen of 16 error structures predate 2006 wwPDB remediation (Mann–Whitney $p = 1.14 \times 10^{-4}$ on the expanded set); resolution does not predict errors ($p = 0.19$).

3.2. MolProbity calibration

On $n = 362$ structures, ChiralFold Ramachandran outlier percentages agree with wwPDB (Spearman $\rho = 0.52$, 95% CI [0.42, 0.61], $p = 1.5 \times 10^{-26}$; mean 0.78% vs 0.83%; Fig. 2). Pearson r is outlier-sensitive (0.33 for $n = 363$; 0.85 after excluding the sole non-protein entry 5M2K); Spearman ρ is the primary agreement metric. ChiralFold adds D-residue chirality auditing absent from MolProbity.

3.3. Formal obstruction and AF3

Lean 4 proofs establish that any chirality readout factoring only through pairwise distances cannot determine sign (Supplementary Table S2); this is a structural limitation of distance-only representations, not a complete model of SE(3)-equivariant predictors. On 41 D-peptide sequences from Childs et al. (2025), ChiralFold yields 0/478 violations by explicit SMILES construction (construction-vs-learning contrast, not fold prediction). A synthetic AF3-mimetic resource benchmark further shows 100% detection/correction recall, 0% residual violations, and a 0.0367 s local correction pipeline across three Childs-system peptides (Supplementary Fig. S4), supporting

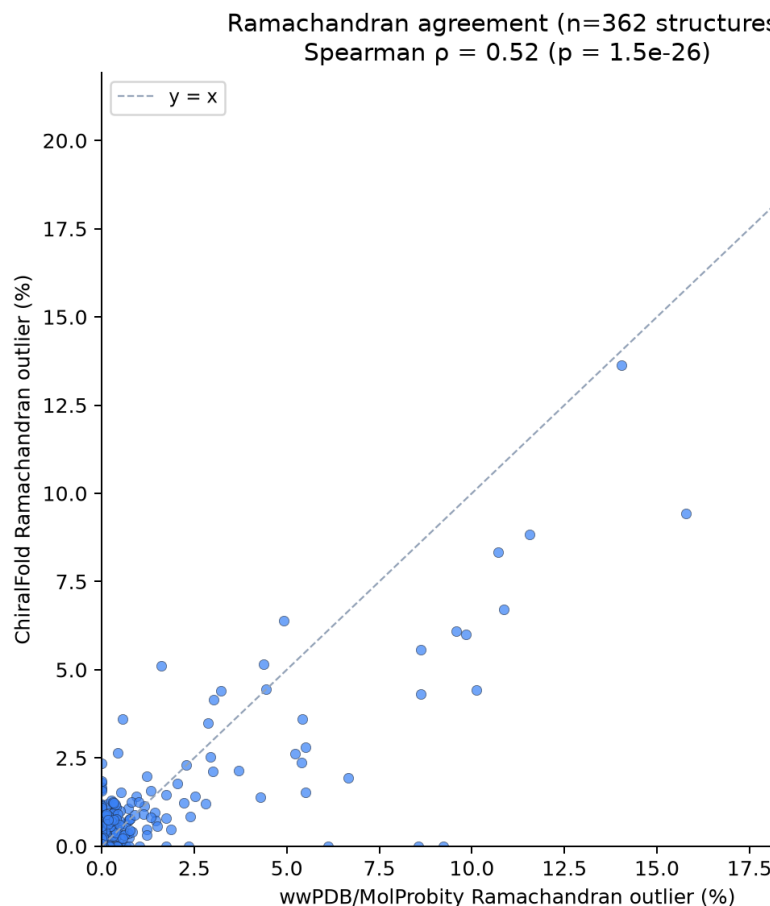


Figure 2 ChiralFold vs wwPDB Ramachandran outlier % ($n = 362$ era-representative structures).

correction as lightweight stereochemistry post-processing rather than AF3 re-inference.

4. Discussion

The signed $C\alpha$ volume distinguishes stereochemistry errors, CCD misassignments, and mislabels without biological context. The 1HHZ case (0.99 Å resolution, D-Ala with $V = +2.70$) shows errors are labeling failures, not density limitations. CCD misassignments are fixable by comparing deposited coordinates to the InChI /m flag at deposition (Shao et al., 2015; Berman et al., 2016).

ChiralFold is a stereochemistry post-processor, not a de novo folder: use AF3 or ESMFold for folding, then audit or correct with ChiralFold. Limitations include mmCIF-only exclusions (245 structures), chi1-only rotamer scoring, single-investigator error subtyping from metadata and literature, two borderline cases with signed volume $< 1.0 \text{ Å}^3$, and no experimental binding validation of mirror-designed peptides.

5. Conclusion

ChiralFold identifies D-residue annotation errors invisible to existing pipelines and motivates a one-line deposition check: compare signed $C\alpha$ volume to the selected CCD stereochemistry

flag. Code, data, formal proofs, and reproduction scripts are publicly available (v3.4.0).

Data availability

Verification data: https://github.com/Tommaso-R-Marena/ChiralFold/blob/master/results/d_residue_verification.csv. Benchmark outputs: [results/ramachandran_279struct_chainfix_41](#). Software: PyPI [chiralfold](#), Docker [ghcr.io/tommaso-r-marena/chiralfold:3.4.0](#), GitHub release [v3.4.0](#).

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Conflict of interest

None declared.

Author contributions

T.R.M. conceived the study, implemented ChiralFold, performed all analyses, formalized proofs, and wrote the manuscript.

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